<u>CECN801 Midterm Exam</u> : Mark the correct answer on your scantron form.

Hand in the scantron form and Bonus question page.

(20 Questions: 5% each)

First Name _____

Last Name _____

Student # _____

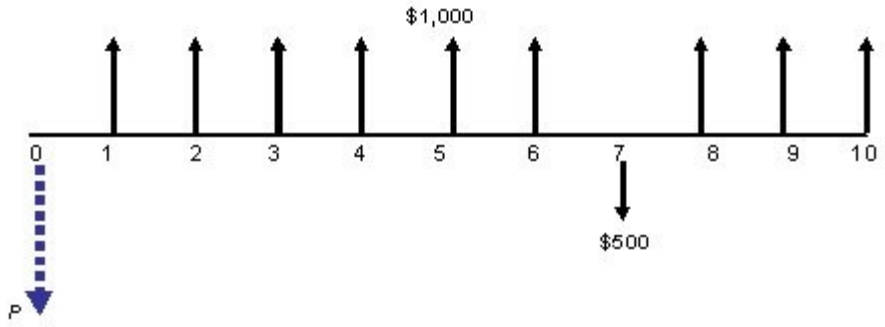
1) In 2010, a biotechnology firm, DNA Map Inc., had $700 million of assets and $280 million of liabilities. Earnings before interest and taxes were $215 million, interest expense was $10 million, and the tax rate was 32%. (Interest expense NOT tax exempt)
ROE = _____

   a) .49
   b) .36
   c) .35
   d) .33
   e) .31

2) How much do you need to invest in equal annual amounts (at year-end) for the next 10 years if you want to withdraw $5,000 at the end of the eleventh year and increase the annual withdrawal by $1,000 each year thereafter until the end of the 25th year? The interest rate is 6%, compounded annually.

   a) 7950.77
   b) 8050.77
   c) 8150.77
   d) 8250.77
   e) 8350.77

3) To withdraw the following $1,000 payment series, determine the absolute minimum amount of deposit (P) that should be made now, assuming that deposits earn an interest rate of 10%, compounded annually. Note that you are making another deposit at the end of year 7 in the amount of $500.



a) 4974.80

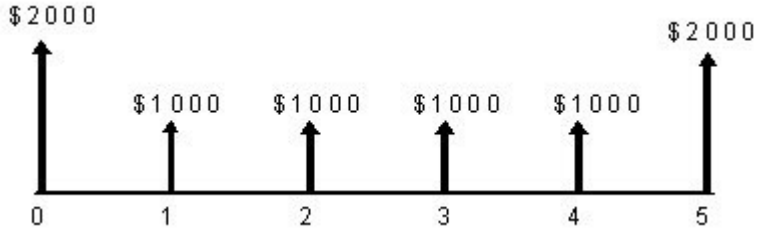
b) 5074.80

c) 5174.80

d) 5274.80

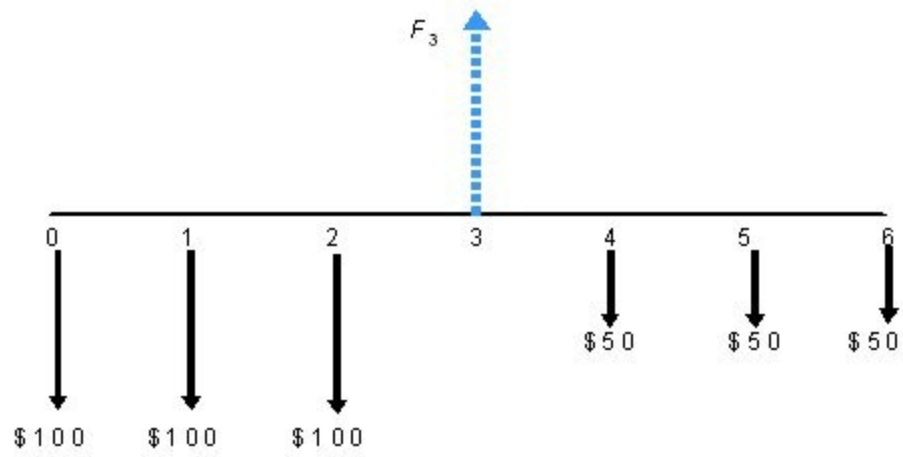
e) 5374.80

4) You want to find the equivalent present worth for the following cash flow series at an interest rate of 15%. Which of the following statements is *incorrect*?



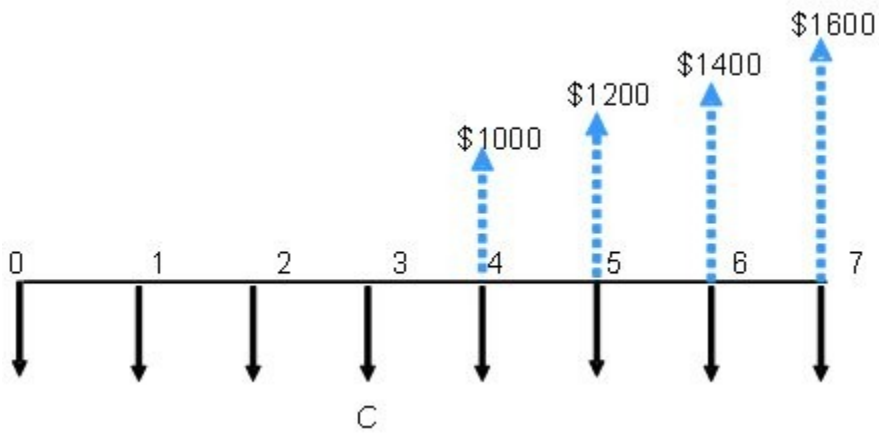
(a) $1,000($P/A$, 15%, 4) + $2,000 + $2,000($P/F$, 15%, 5)
(b) $1,000($P/F$, 15%, 5) + $1,000($P/A$, 15%, 5) + $2,000
(c) [$1,000($F/A$, 15%, 5) + $1,000] ($P/F$, 15%, 5) + $2,000
(d) [$1,000($F/A$, 15%, 4) + $2,000] ($P/F$, 15%, 4) + $2,000
(e) {$1,000($P/A$, 15%, 6) + $1,000[($P/F$,15%,1) + ($P/F$,15%,6)]}($F/P$,15%,1)

5) What value of F_3 would be equivalent to the payments shown in the cash flow diagram below? Assume the interest rate is 10%, compounded annually.



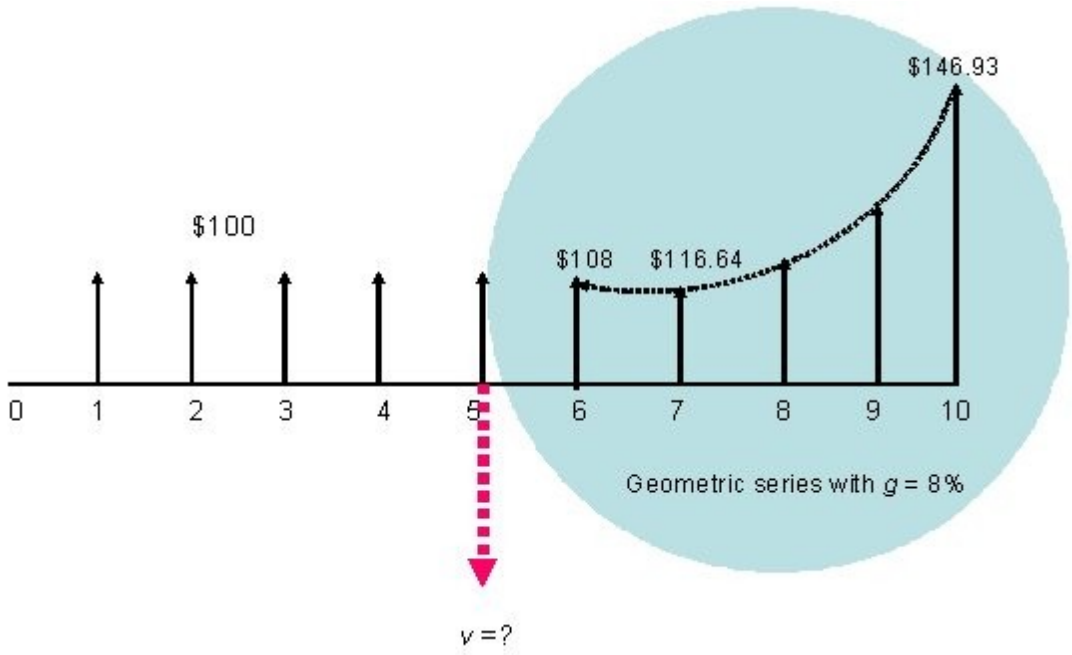
a) $488.45
b) $478.45
c) $468.45
d) $458.45
e) $448.45

6) Consider the cash flow series shown below. Determine the required annual deposits (time zero PLUS remainder are end of year) that will generate the cash flows from years 4 to 7. Assume the interest rate is 10%, compounded annually.



a) 497.93
b) 507.93
c) 517.93
d) 527.93
e) 537.93

7) Compute the value of V (V_5) in the following cash flow diagram. Assume i = 12%.

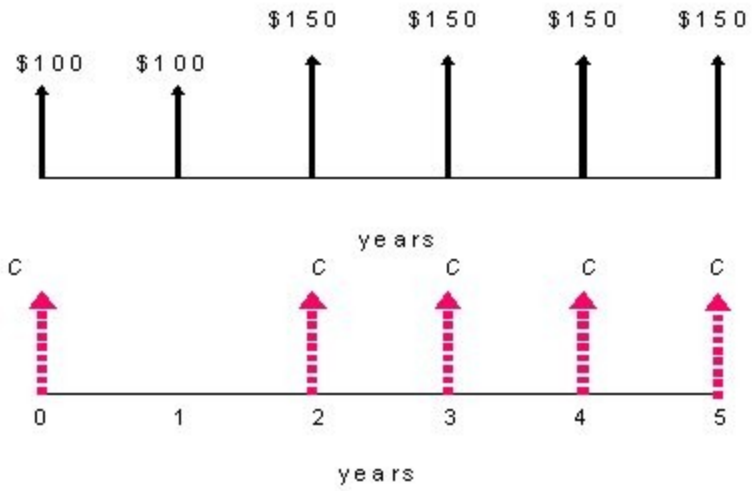


a) $1044.19
b) $1054.19
c) $1064.19
d) $1074.19
e) $1084.19

8) If you deposit $1,000 and leave it to accumulate interest over three years at 9% annual interest, what is the interest earned in the 2nd year?

a) $188.1
b) $158.1
c) $128.1
d) $98.1
e) It depends on the amount of the 3rd year withdrawal.

9) What value of C makes the following two annual cash flows equivalent at an annual rate of 10%?



a) 155.54

b) 160.54

c) 165.54

d) 170.54

e) 175.54

10) Today is your birthday and you decide to start saving for your retirement. You will retire on your 65th birthday and will need $50,000 per year at the **end** of each of following 20 years (after your 65th birthday). You will make a first deposit one year from today (on your next birthday) in an account paying 8% interest annually and continue to make an equal amount of deposit each year up to the year before you plan to retire (your final deposit will be on the day of your 65th birthday). If an annual deposit of $6,715 will allow you to reach your goal, what 'birthday' are you celebrating today?

a) 25

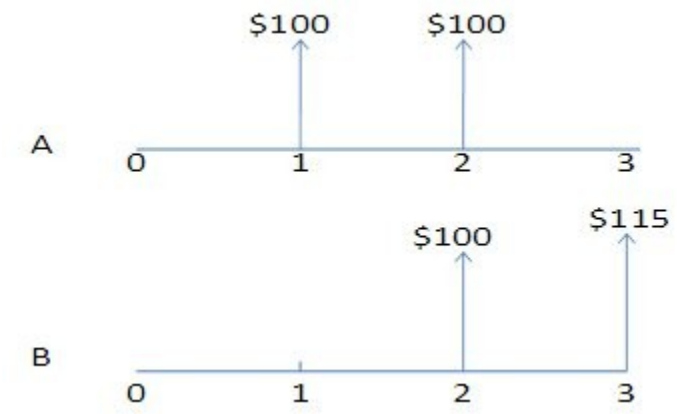
b) 45

c) 40

d) 30

e) 35

11) Consider the following two cash flow transactions that are said to be economically equivalent at an interest rate of *i*. Determine the interest rate that establishes such an economic equivalence between the two cash flows.



a) 7.638%
b) 7.438%
c) 7.238%
d) 7.038%
e) 6.838%

12) Your starting salary as a mechanical engineer is $55,000 (this year). A total of 10% of your salary each year will be placed in the mutual fund of your choice (at the end of each year). You can also count on a 5% annual salary increase for the next 30 years of employment (starting next year). If the mutual fund will average 9% annual return over the course of your career, how much can you expect at retirement?

a) $92,709.41
b) $1,000,020.19
c) $457,757.45
d) $157,235.91
e) $1,230,040.71

13) A couple is planning to finance their 5–year–old daughter's college education. They established a college fund that earns 9.5%, compounded annually. What annual deposit must be made from the daughter's 5th birthday (now) up to and including her 16th birthday to meet the future college expenses as follows: (Assume that today is her 5th birthday and deposits will be made on the morning of her birthdays and will accumulate interest over the year).

18th Bday: 25000

19th Bday: 27000

20th Bday: 29000

21st Bday: 31000

a) 3916.64
b) 4116.64
c) 4316.64
d) 4516.64
e) 4716.64

14. Your bank calculates the interest based on 12.5% APR (monthly compounding) on your credit card balance. Suppose that your current outstanding balance is $2,000 and you skip payments for two months. What would be the total balance two months from now?

a) $2041.88
b) $2051.88
c) $2061.88
d) $2071.88
e) $2091.88

15. At what rate of interest (APR compounded monthly) will an investment quintuple itself (grow to five times its present value) in 5 years?

a) 22.62%
b) 24.62%
c) 26.62%
d) 29.62%
e) 32.62%

16. If the interest rate is 9.5% compounded continuously, what is the required uniform quarterly payment (end of quarter payments) to repay a loan of $15,000 in four years?

a) $1040.33
b) $1080.33
c) $1110.33
d) $1140.33
e) $1170.33

17. A building is priced at $100,000. If you make a down payment in the amount of $30,000 and wish to make $1,000 monthly (end of the month) payments, how many months will it take to pay off the loan? Interest is charged at a rate of 12%, compounded monthly. (Hint: rearrange the formula so there is only one term with an *N* appearing.)

a) 111
b) 121
c) 131
d) 141
e) 151

18. Two financing options are being offered in purchasing a vehicle:

- Option 1: cash purchase at $20,500.
- Option 2: Debt purchase – Purchase the vehicle at the price of $22,600 and finance it over three years with equal monthly payments at 3.5% APR compounded monthly.

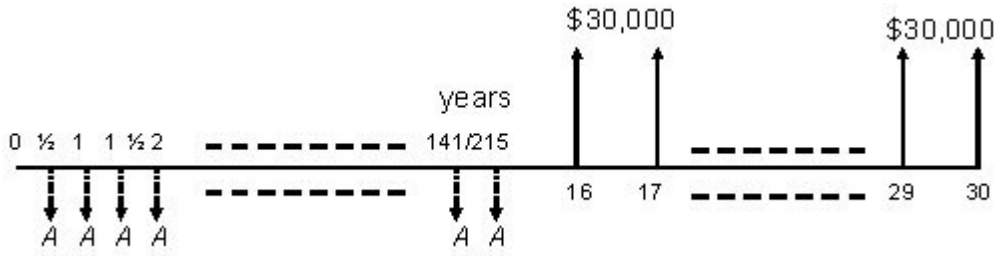
By how much (in Present Worth) is Option 1 more economical at an interest rate of 6% APR compounded monthly? (Suggestion: round to five **positive** decimal places)

a) $1268.16
b) $1368.16
c) $1468.16
d) $1568.16
e) $1668.16

19. What is the equal–payment series for 8 years (end of year payments) that is equivalent to a payment series starting with $15,000 at the end of the first year and decreasing by $2,000 each year over the next seven years? Interest is 9% compounded monthly.

a) $8833.40
b) $8933.40
c) $9033.40
d) $9133.40
e) $9233.40

20. Henry Jones is planning to retire in 15 years. He wishes to deposit an equal amount (A) every 6 months until he retires so that, beginning one year following his retirement, he will receive annual payments of $30,000 for the next 15 years. Determine the value of ***A*** which he should deposit every 6 months if the interest rate is 8%, compounded quarterly.



a) $4081.27
b) $4181.27
c) $4281.27
d) $4381.27
e) $4481.27